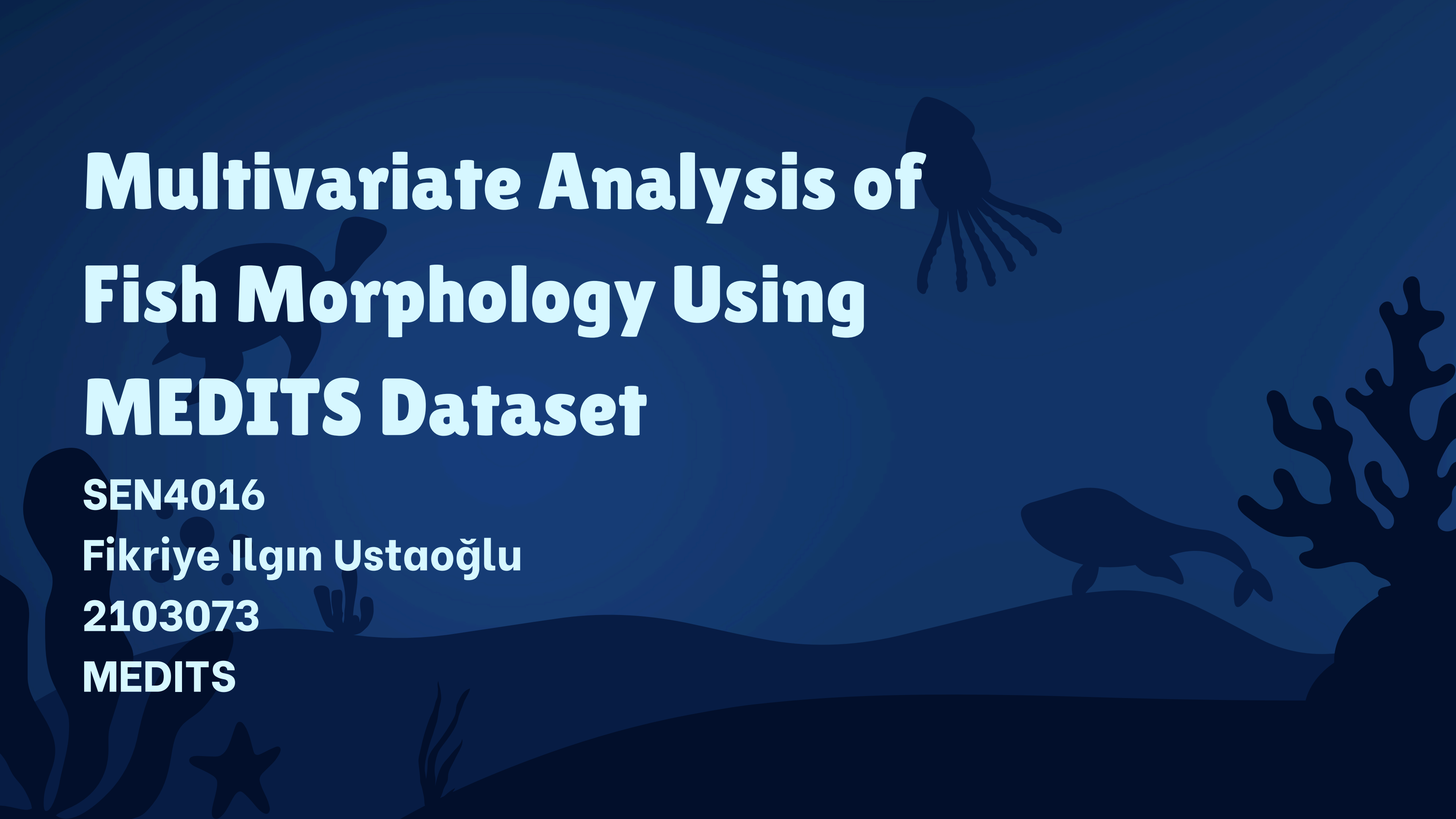


Multivariate Analysis of Fish Morphology Using MEDITS Dataset



SEN4016

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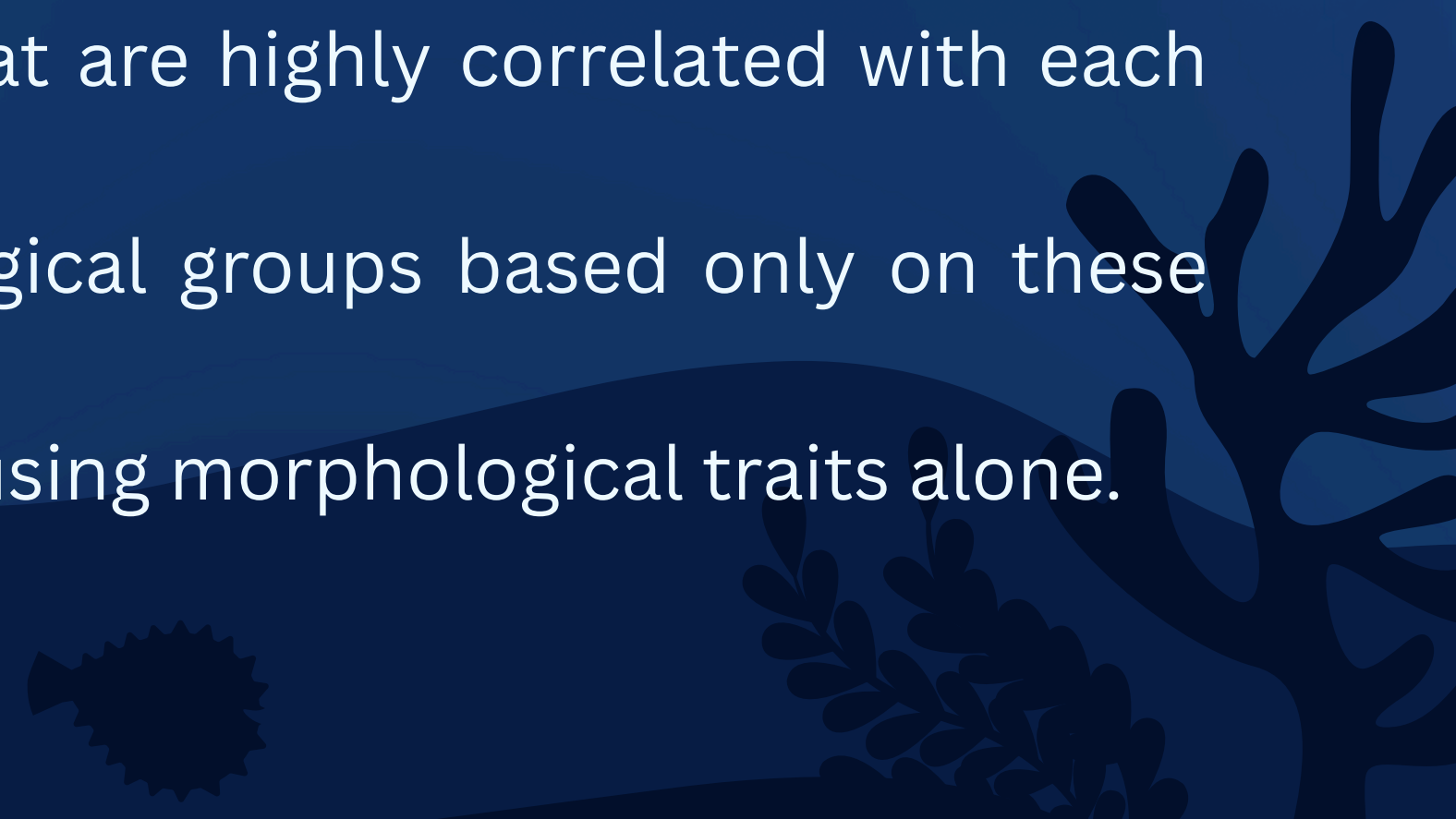
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MEDITS

What is MEDITS?

MEDITS (Mediterranean International Trawl Survey) is a large-scale scientific survey program conducted in the Mediterranean Sea. It collects standardized biological and morphological data from fish species to support marine research and stock assessment. In this study, morphological measurements from 89 fish species and over 1,500 individual observations are analyzed.

What are The Problems of MEDITS?

- The dataset has many morphological variables that are highly correlated with each other.
 - It is unclear whether fish form natural morphological groups based only on these measurements.
 - It is challenging to accurately classify fish species using morphological traits alone.
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Applied Methodologies

- Principal Component Analysis (PCA) was applied to reduce dimensionality and handle correlated morphological variables.
- Cluster Analysis was used to identify natural morphological groups without using species labels.
- Discriminant Analysis was performed to evaluate how well morphological traits can classify fish species.

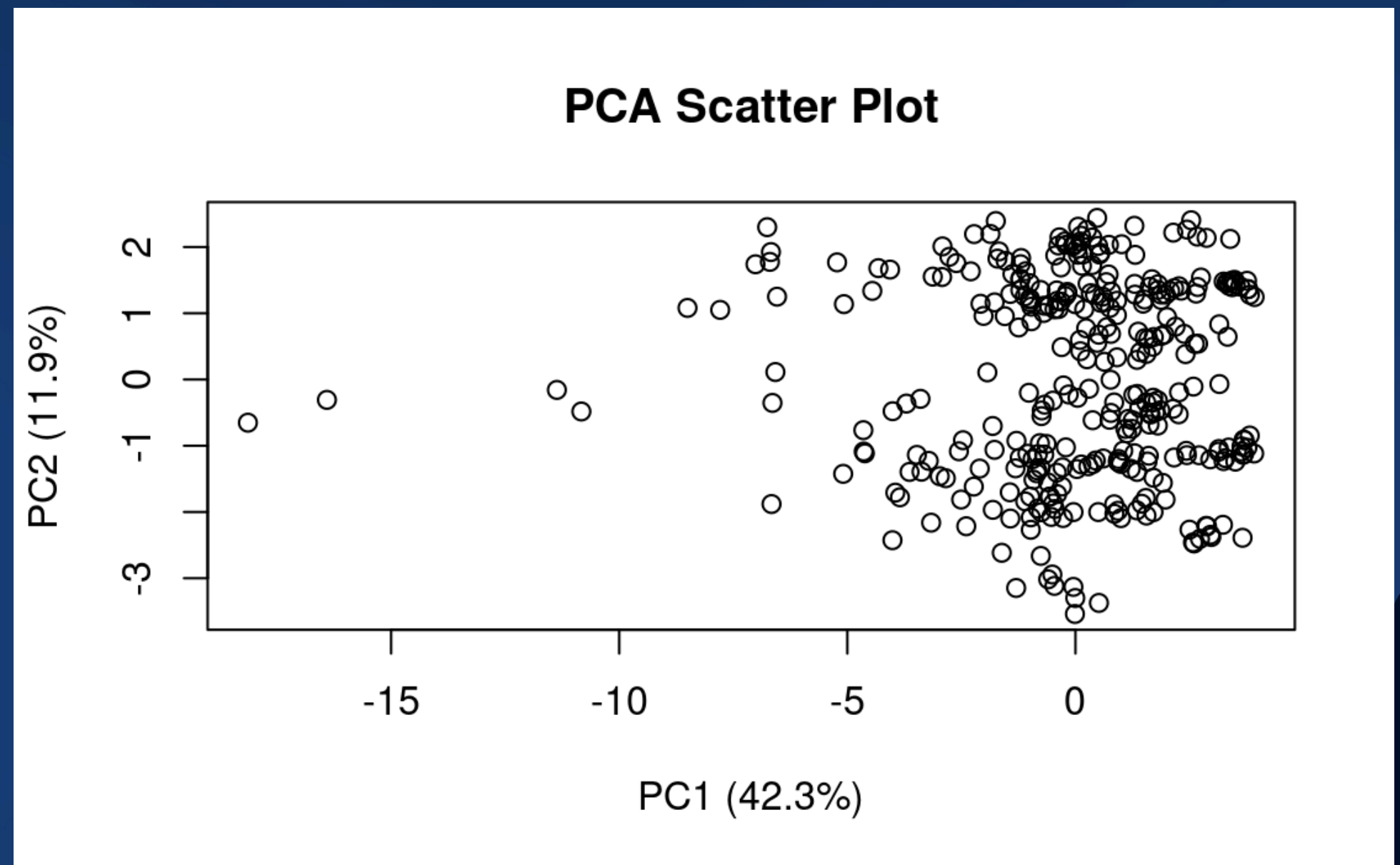


In this order:

PCA → Clustering → Discriminant Analysis

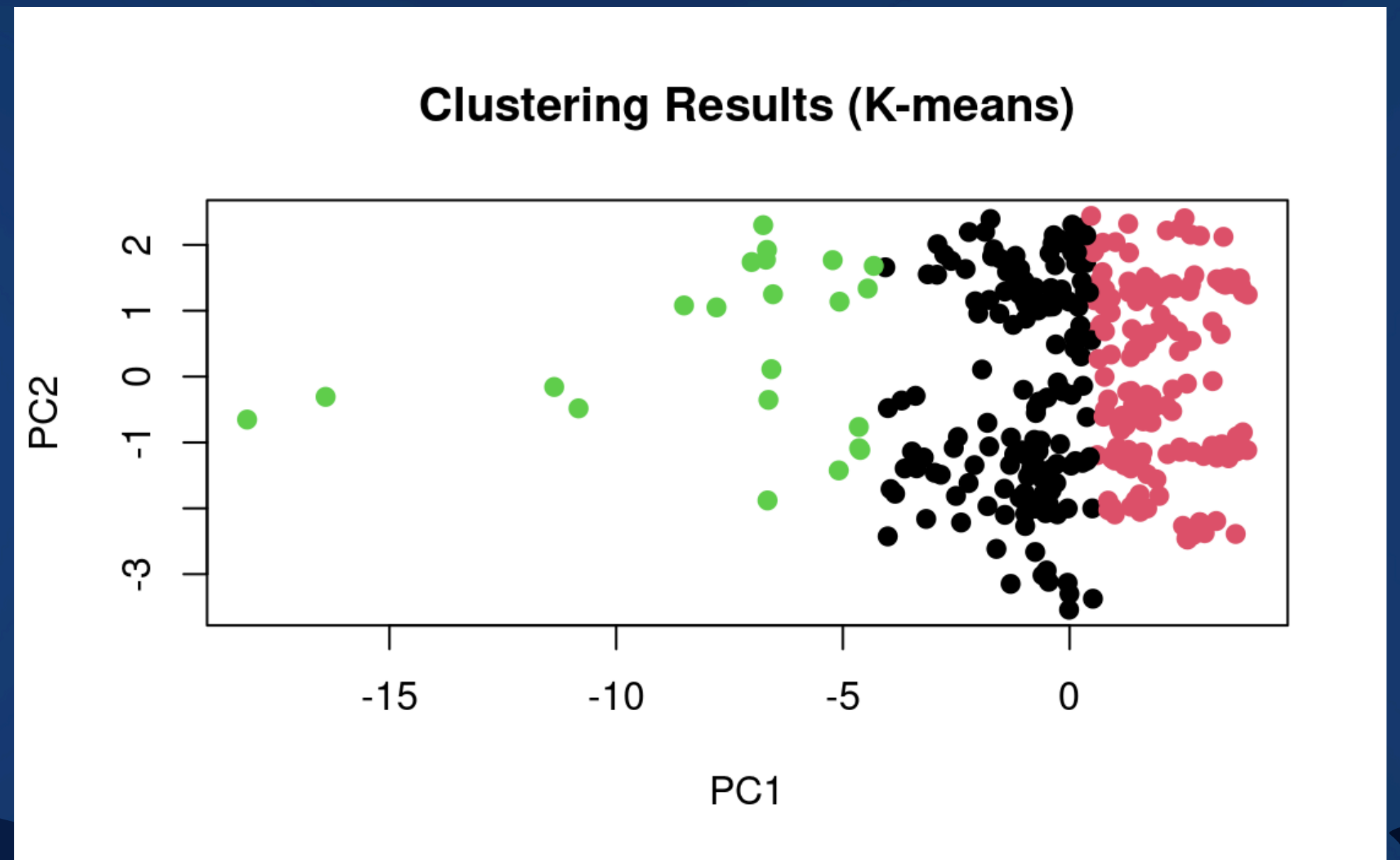
Results of PCA

- PCA reduced the high-dimensional morphological data into a smaller number of principal components.
- The first two principal components explained a large proportion of the total variance.
- These components capture the main patterns of morphological variation among fish.



Results of Clustering

- Cluster analysis was applied to identify natural morphological groupings.
- The results show partially separated clusters based on morphological similarity.
- This suggests that fish morphology exhibits structured patterns independent of species labels.



Results of Discriminant Analysis

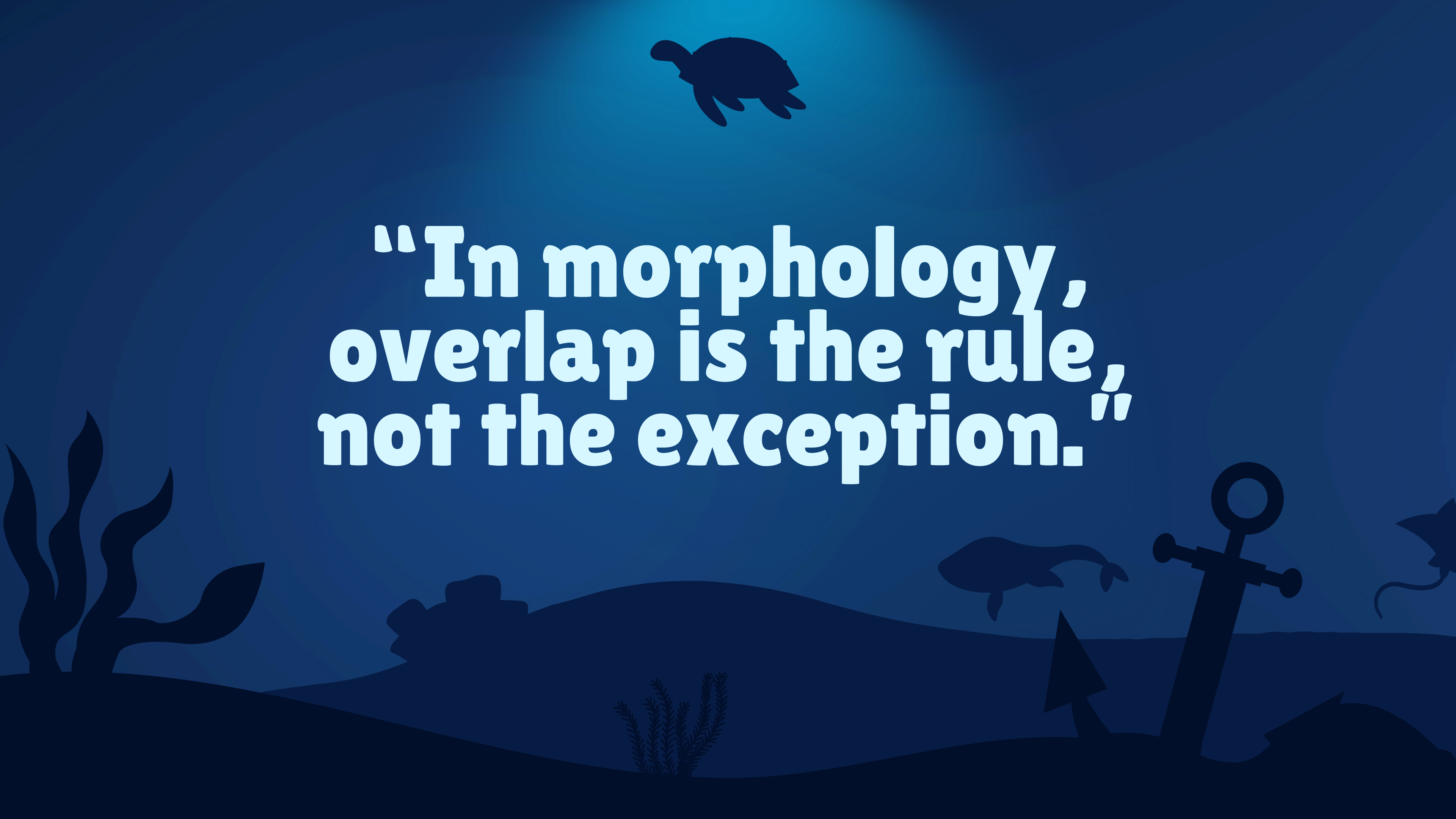
- Discriminant analysis was applied to classify fish species using morphological traits.
- The model achieved a classification accuracy of approximately 47%.
- The moderate accuracy indicates strong morphological similarity among many species.



Conclusion

- The MEDITS dataset contains high-dimensional and correlated morphological measurements.
- PCA successfully reduced dimensionality while preserving the main patterns of variation.
- Clustering analysis suggested partial natural grouping among fish based on morphology.
- Discriminant analysis achieved moderate classification accuracy, indicating limitations of morphology-only classification.



The background is a dark blue gradient with silhouettes of various underwater elements. At the top center is a turtle swimming. In the middle right is a fish swimming. At the bottom right is a large anchor. On the left side are seaweed-like plants. At the bottom center is a small, feathery plant. The text is centered in the upper half of the image.

**"In morphology,
overlap is the rule,
not the exception."**